



PROGRAM NAME B.Tech

YEAR/SEMESTER ~~4th~~ 417

FIRST CONTINUOUS ASSESSMENT, 2025-26

Course Name Datawarehouse and Data Mining Course Code CS-418

Time : 1.30 hrs.

Max. Marks : 20

- Note : (i) The question paper contains **Three** Sections.
(ii) Section-A is compulsory, Section-B and C contains internal choice.

SECTION-A

(MCQS/FILL UP THE BLANKS/VERY SHORT ANSWER QUESTIONS)

1. Attempt **ALL** parts of the following questions: 0.5 x 10 = 5
- (a) The data Warehouse is _____.
A. read only. B. write only. C. read write only. D. none. [BT-1, CO-1, PO-1]
- (b) The data is stored, retrieved & updated in _____. [BT-2, CO-1, PO-2]
- (c) _____ describes the data contained in the data warehouse. [BT-2, CO-1, PO-2]
- (d) _____ is the specialized data warehouse database. [BT-1, CO-1, PO-1]
- (e) _____ defines the structure of the data held in operational databases and used by operational applications. [BT-2, CO-2, PO-3]
- (f) Query tool is meant for _____. [BT-1, CO-2, PO-5]
- (g) Record cannot be updated in _____. [BT-1, CO-1, PO-2]
- (h) Data warehouse contains _____ data that is never found in the operational environment. [BT-2, CO-1, PO-1]
- (i) Data scrubbing is _____. [BT-2, CO-2, PO-5]
- (j) _____ are responsible for running queries and reports against data warehouse tables. [BT-2, CO-1, PO-5]

SECTION-B

(SHORT ANSWER QUESTIONS)

2. Attempt any **TWO** of the following questions: 2.5 x 2 = 5
- (a) List two reasons for the failures of past decision support systems. [BT-1, CO-1, PO-2]
- (b) Differentiate between Star Schema and Snowflake Schema. [BT-2, CO-2, PO-3]
- (c) Suppose a bank wants to analyze customer transactions over time—why is a Data Warehouse more suitable than an operational DBMS? [BT-3, CO-1, PO-12]
- (d) Imagine a healthcare Data Warehouse: how can poor data quality affect decision making? [BT-3, CO-2, PO-2]

SECTION-C

(LONG ANSWER QUESTIONS)

3. Attempt any **TWO** of the following questions: 5 x 2 = 10
- (a) Compare and contrast the architecture and functionalities of OLTP systems and Data Warehouses with suitable examples. [BT-4, CO-1 PO-2]
- (b) ETL is often considered the most critical step in Data Warehousing. Critically examine the challenges of ETL and propose strategies to overcome them. [BT-5, CO-2, PO-5]
- (c) Assume you are designing a retail analytics system: justify why OLAP is essential, and explain which OLAP features would be most beneficial. [BT-5, CO-1, PO-12]
- (d) Illustrate with an example how OLAP operations (roll-up, drill-down, slice, dice) can support managerial decision making in a banking system. [BT-4, CO-2, PO-5]